

2023 Ph H1 Q22

Section: Electricity

Topic: Capacitors

Question Summary

One coulomb per volt is equivalent to one ... (choose the correct unit).

Worked Solution

Definition of capacitance: $C = Q / V$.

So the unit of capacitance is coulomb per volt.

Therefore, one coulomb per volt equals one farad.

Final Answer: B

Revision Tips

- Capacitance $C = Q/V \rightarrow$ units: farad (F) = coulomb per volt (C/V).
- Energy stored: $E = \frac{1}{2} C V^2$ (units joule).
- Charge-voltage is linear for ideal capacitors (on d.c. steady-state).